

With or without replacement



- 1** A pile of playing cards has 4 diamonds and 3 hearts. A second pile has 2 diamonds and 5 hearts. One card is selected at random from the first pile, then the second.
 - a** Construct a probability tree of this situation with the correct probability on each branch.
 - b** Find the probability of selecting two hearts.

- 2** In tennis if the first serve is a fault (out or in the net), the player takes a second serve. A player serves with the following probabilities:
 - First serve in: 0.55
 - Second serve in: 0.81
 - a** Construct a probability tree showing the probability of the first two serves either being in or a fault.
 - b** Find the probability that the player needs to make a second serve.
 - c** Find the probability that the player makes a double fault (both serves are a fault).

- 3** A container holds four counters coloured red, blue, green and yellow. Draw a tree diagram representing all possible outcomes when two draws are done, and the first counter is:
 - a** Replaced before the second draw.
 - b** Not replaced before the next draw.

-  **4** A container holds three cards coloured red, blue and green.
 - a** Draw a tree diagram representing all possible outcomes when two draws occur, and the card is not replaced before the next draw.
 - b** Find the probability of drawing the blue card first.
 - c** Find the probability of drawing a blue card in either the first or second draw.
 - d** Find the probability of drawing at least one blue card if the cards are replaced before the next draw.

- 5** Consider the word WOLLONGONG. If three letters are randomly selected from it without replacement, find the probability that:
 - a** The letters are W, O, L, in that order.
 - b** The letters are O, N, G, in that order.
 - c** All three letters are O.
 - d** None of the three letters is an O.



- 6** Two marbles are randomly drawn without replacement from a bag containing 1 blue, 2 red and 3 yellow marbles.
- Construct a tree diagram to show the sample space.
 - Find the probability of drawing the following:
 - A blue marble and a yellow marble, in that order.
 - A red marble and a blue marble, in that order.
 - 2 red marbles.
 - No yellow marbles.
 - 2 blue marbles.
 - A yellow marble and a red marble, in that order.
 - A yellow and a red marble, in any order.
- 7** There are 4 green counters and 8 purple counters in a bag. Find the probability of choosing a green counter, not replacing it, then choosing a purple counter.
- 8** A standard deck of cards is used and 3 cards are drawn out. Find the probability, in fraction form, of the following:
- All 3 cards are diamonds if the cards are drawn with replacement.
 - All 3 cards are diamonds if the cards are drawn without replacement.
- 9** A number game uses a basket with 6 balls, all labelled with numbers from 1 to 6. 2 balls are drawn at random. Find the probability that the ball labelled 3 is picked once if the balls are drawn:
- With replacement.
 - Without replacement.
- 10** A hand contains a 10, a jack, a queen, a king and an ace. Two cards are drawn from the hand at random, in succession and without replacement. Find the probability that:
- The ace is drawn.
 - The king is not drawn.
 - The queen is the second card drawn.
- 11** Eileen randomly selects two cards, with replacement, from a normal deck of cards. Find the probability that:
- The first card is a queen of spades and the second card is a 4 of clubs.
 - The first card is spades and the second card is a 4.
 - The first card is a Queen and the second card is black.
 - The first card is not a 7 and the second card is not Clubs.



- 12** Find the probability of drawing a green counter from a bag of 5 green counters and 6 black counters, replacing it and drawing another green counter.
- 13** A chess player is placed into a draw where in each match he has a 30% chance of winning. Find the probability that:
- a** He wins his first two matches.
 - b** He wins his first three matches.
 - c** He wins his first two matches and then loses his third match.
- 14** Beth randomly selects three cards, with replacement, from a normal deck of cards. Find the probability that:
- a** The cards are queen of diamonds, king of spades, and king of diamonds, in that order.
 - b** The cards are all black.
 - c** The first card is a 9, the second card is a heart and the third card is red.
 - d** The cards are all hearts.
 - e** None of the cards is a 9.
- 15** From a standard pack of cards, one card is randomly drawn and then put back into the pack. A second card is then drawn. Calculate the probability that:
- a** Neither of the cards are diamonds
 - b** At least one of the cards is a diamond
- 16** Three marbles are randomly drawn with replacement from a bag containing 6 red, 4 yellow, 3 white, 2 black and 4 green marbles. Find the probability of drawing:
- a** Three white marbles
 - b** No green marbles
 - c** At least one red marble
 - d** At least one white marble
- 17** Amelia randomly selects two cards, with replacement, from a normal deck of cards. Find the probability that:
- a** Both cards are red
 - b** Both cards are the same colour
 - c** Both cards are of different colours

18 A bag contains four marbles - red, green, blue and yellow. Beth randomly selects a marble, returns the marble to the bag and selects another marble.



- a** Construct a tree diagram for the experiment given.
- b** Find the probability of Beth selecting:
 - i** A blue and a yellow marble.
 - ii** A blue followed by a yellow marble.
 - iii** 2 red marbles.
 - iv** 2 marbles of the same colour.
 - v** 2 marbles of different colours.

19 James, a test cricketer, analysed his past innings and found his probabilities of scoring particular numbers of runs, as shown in the table:

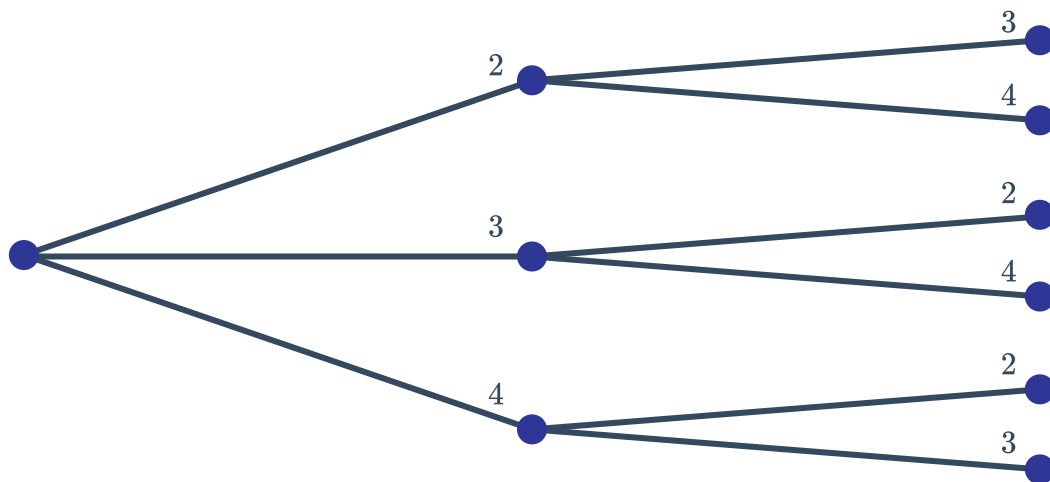
Number of runs	0	1 – 20	21 – 49	50 – 99	100+
Probability	$\frac{1}{10}$	$\frac{3}{10}$	$\frac{3}{10}$	$\frac{2}{10}$	$\frac{1}{10}$

If James is selected to play in the next test match, calculate the probability that he scores:

- a** 0 runs in exactly one of the two innings.
- b** 0 runs in both innings.
- c** At least 50 runs in at least one of the innings.
- d** A total of 100 or more runs in each of the two innings.
- e** More than 20 runs in only one of the two innings.



- 20** Three cards labeled 1, 2, 3 are placed face down on a table. Two of the cards are selected randomly to form a two-digit number. The possible outcomes are displayed in the following probability tree:



- a** List the sample space of two digit numbers produced by this process.
 - b** Find the probability that 2 is a digit in the number.
 - c** Find the probability that the sum of the two selected cards is even.
 - d** Find the probability of forming a number greater than 40.
- 21** In a school, 25% of students ride skateboards and 20% of students have dark hair. One student is selected at random. Find the probability that the student:
- a** Has dark hair and rides a skateboard.
 - b** Has light hair and does not ride a skateboard.
 - c** Has dark hair and does not ride a skateboard.
 - d** Has light hair and rides a skateboard.
- 22** The ratio of left-handed people to right-handed people in a country is 4 : 3. Two people are surveyed at random. Calculate the probability that:
- a** Both people are left-handed.
 - b** One person is left-handed and the other is right-handed.
 - c** At least one person is right-handed.